

Theory Note for `orth_cqkud.py`: Overlap-Orthonormal Chebyshev Quantum Krylov Diagonalization

QCANT generated theory note

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Abstract

The generated `orth_cqkud.py` module implements a Chebyshev-polynomial quantum Krylov method and returns an overlap-orthonormalized basis. The mathematical justification is the standard Rayleigh–Ritz projection principle applied to a nonorthogonal Krylov basis, followed by Lowdin symmetric orthonormalization of the overlap matrix. This note explains why the Chebyshev recurrence spans a Krylov subspace, why the raw basis leads to the generalized eigenvalue problem $\mathbf{H}_K \mathbf{c} = E \mathbf{S} \mathbf{c}$, and why the transformation $\mathbf{X} = \mathbf{U}_r \mathbf{\Lambda}_r^{-1/2}$ converts that problem into an ordinary Hermitian eigenvalue problem in an orthonormal basis.

1 Purpose of the orthonormal CQKUD variant

The original CQKUD construction generates vectors

$$|\chi_0\rangle = |\psi_0\rangle, \quad |\chi_1\rangle = \tilde{H} |\psi_0\rangle, \quad |\chi_{n+1}\rangle = 2\tilde{H} |\chi_n\rangle - |\chi_{n-1}\rangle,$$

where

$$\tilde{H} = \frac{H - \eta I}{\alpha}.$$

These vectors are useful Krylov trial states, but they are generally not mutually orthogonal. The generated file `orth_cqkud.py` keeps the same Chebyshev construction, then forms orthonormal linear combinations before returning `basis.states`. This makes the returned CQKUD basis behave more like a qscEOM determinant/configuration basis: the basis vectors returned to the user satisfy

$$\langle q_a | q_b \rangle \approx \delta_{ab}.$$

The key implementation block is:

$$\begin{aligned} \mathbf{S} &= \mathbf{B}^* \mathbf{B}^T, \\ \mathbf{H}_K &= \mathbf{B}^* \mathbf{H} \mathbf{B}^T, \\ \mathbf{S} &= \mathbf{U} \mathbf{\Lambda} \mathbf{U}^\dagger, \\ \mathbf{X} &= \mathbf{U}_r \mathbf{\Lambda}_r^{-1/2}, \\ \mathbf{Q} &= \mathbf{X}^T \mathbf{B}, \\ \mathbf{H}_{\text{ortho}} &= \mathbf{X}^\dagger \mathbf{H}_K \mathbf{X}. \end{aligned}$$

Here $\mathbf{B}$ stores the raw Chebyshev vectors as rows, $\mathbf{Q}$ stores the returned orthonormal vectors as rows, and the subscript r denotes that only overlap eigenvalues larger than `overlap_tol` are kept.

2 Chebyshev Krylov basis

Let H be a Hermitian electronic Hamiltonian and let $|\psi_0\rangle$ be a normalized initial state, often the Hartree–Fock determinant. CQKUD first shifts and scales the Hamiltonian:

$$\tilde{H} = \frac{H - \eta I}{\alpha}.$$

If $E_{\min}$ and $E_{\max}$ bound the spectrum of H , the common choice

$$\eta = \frac{E_{\max} + E_{\min}}{2}, \quad \alpha = \frac{E_{\max} - E_{\min}}{2}$$

maps the spectrum of H into approximately $[-1, 1]$. This interval is the natural stability interval for Chebyshev polynomials $T_n(x)$, which obey

$$T_0(x) = 1, \quad T_1(x) = x, \quad T_{n+1}(x) = 2xT_n(x) - T_{n-1}(x).$$

The generated vectors therefore satisfy

$$|\chi_n\rangle = T_n(\tilde{H}) |\psi_0\rangle.$$

2.1 Equivalence to a polynomial Krylov space

The degree of $T_n(x)$ is n , and its leading coefficient is nonzero. Consequently,

$$\text{span}\{T_0(\tilde{H}) |\psi_0\rangle, \dots, T_m(\tilde{H}) |\psi_0\rangle\} = \text{span}\{|\psi_0\rangle, \tilde{H} |\psi_0\rangle, \dots, \tilde{H}^m |\psi_0\rangle\}.$$

Because $\tilde{H}$ is an affine transformation of H with $\alpha \neq 0$, this is the same polynomial Krylov subspace as

$$\mathcal{K}_{m+1}(H, |\psi_0\rangle) = \text{span}\{|\psi_0\rangle, H |\psi_0\rangle, \dots, H^m |\psi_0\rangle\}.$$

Thus the Chebyshev recurrence changes the polynomial basis used to represent the subspace; it does not change the underlying Krylov subspace, up to numerical rank effects.

3 Rayleigh–Ritz projection with a nonorthogonal basis

Let the raw CQKUD basis contain m vectors

$$\{|\chi_0\rangle, |\chi_1\rangle, \dots, |\chi_{m-1}\rangle\}.$$

An approximate eigenvector in this subspace is

$$|\Psi(c)\rangle = \sum_{j=0}^{m-1} c_j |\chi_j\rangle.$$

The Rayleigh quotient is

$$R(c) = \frac{\langle \Psi(c) | H | \Psi(c) \rangle}{\langle \Psi(c) | \Psi(c) \rangle} = \frac{c^\dagger \mathbf{H}_K c}{c^\dagger \mathbf{S} c},$$

where

$$(\mathbf{S})_{ij} = \langle \chi_i | \chi_j \rangle, \quad (\mathbf{H}_K)_{ij} = \langle \chi_i | H | \chi_j \rangle.$$

Stationarity of $R(c)$ gives the generalized Hermitian eigenproblem

$$\mathbf{H}_K c = E \mathbf{S} c.$$

This is the standard nonorthogonal-basis Rayleigh–Ritz equation. It is exactly what the raw CQKUD construction requires before orthonormalization.

4 Overlap orthonormalization

The raw overlap matrix $\mathbf{S}$ is Hermitian positive semidefinite:

$$c^\dagger \mathbf{S} c = \left\| \sum_j c_j |\chi_j\rangle \right\|^2 \geq 0.$$

If the vectors are linearly independent, $\mathbf{S}$ is positive definite. In floating-point arithmetic, however, high-order Krylov vectors often become nearly linearly dependent. The generated implementation therefore diagonalizes

$$\mathbf{S} = \mathbf{U} \mathbf{\Lambda} \mathbf{U}^\dagger$$

and keeps only eigenvalues $\lambda_a > \tau$, where $\tau = \text{overlap_tol}$. Let $\mathbf{U}_r$ contain the retained eigenvectors and let $\mathbf{\Lambda}_r$ be the retained diagonal eigenvalue matrix. Define

$$\mathbf{X} = \mathbf{U}_r \mathbf{\Lambda}_r^{-1/2}.$$

The orthonormalized basis vectors are

$$|q_a\rangle = \sum_i X_{ia} |\chi_i\rangle.$$

In the row-vector convention used by the code, this is

$$\mathbf{Q} = \mathbf{X}^T \mathbf{B}.$$

4.1 Proof of orthonormality

The overlap between two returned vectors is

$$\langle q_a | q_b \rangle = \sum_{ij} X_{ia}^* \langle \chi_i | \chi_j \rangle X_{jb} = (\mathbf{X}^\dagger \mathbf{S} \mathbf{X})_{ab}.$$

Using $\mathbf{S} = \mathbf{U}_r \mathbf{\Lambda}_r \mathbf{U}_r^\dagger$ on the retained subspace,

$$\mathbf{X}^\dagger \mathbf{S} \mathbf{X} = \mathbf{\Lambda}_r^{-1/2} \mathbf{U}_r^\dagger \mathbf{U}_r \mathbf{\Lambda}_r \mathbf{U}_r^\dagger \mathbf{U}_r \mathbf{\Lambda}_r^{-1/2} = \mathbf{I}_r.$$

Therefore the returned vectors are orthonormal within numerical precision:

$$\langle q_a | q_b \rangle = \delta_{ab}.$$

5 Transforming the generalized eigenproblem

The Hamiltonian matrix in the returned orthonormal basis is

$$(\mathbf{H}_{\text{ortho}})_{ab} = \langle q_a | H | q_b \rangle = (\mathbf{X}^\dagger \mathbf{H}_K \mathbf{X})_{ab}.$$

Since the basis is orthonormal, the generalized problem becomes the ordinary Hermitian eigenvalue problem

$$\mathbf{H}_{\text{ortho}} y = E y.$$

This is why `orth_cqkud.py` returns both $\mathbf{S}_{\text{ortho}} \approx \mathbf{I}$ and $\mathbf{H}_{\text{ortho}}$, and then calls a Hermitian eigensolver on $\mathbf{H}_{\text{ortho}}$. The Ritz energies are invariant under this full-rank change of basis on the retained subspace. If thresholding removes near-null directions, the method solves the variational problem in the numerically stable rank- r Krylov subspace.

6 Why this is consistent with qscEOM-style bases

In qscEOM, the basis configurations are occupation patterns in the computational basis. Distinct computational basis states are orthonormal by construction. CQKUD vectors, by contrast, are generated by repeated polynomial actions of H or $\tilde{H}$ on one initial state; such vectors are not generally orthogonal. The `orth_cqkud.py` post-processing step gives CQKUD the same returned-basis property that qscEOM has naturally:

$$\text{returned overlap matrix} \approx I.$$

The important difference is interpretive: a returned CQKUD vector $|q_a\rangle$ is usually a linear combination of several Chebyshev vectors, not the chronological state $T_a(\tilde{H})|\psi_0\rangle$ itself.

7 Numerical safeguards and limitations

1. **Rank reduction.** If some overlap eigenvalues are below `overlap_tol`, the returned basis has fewer vectors than the requested $n_{\text{steps}} + 1$. This is mathematically appropriate: linearly dependent vectors cannot all be made mutually orthonormal.
2. **Conditioning.** Directly solving $\mathbf{H}_K \mathbf{c} = \mathbf{E} \mathbf{S} \mathbf{c}$ can be unstable when $\mathbf{S}$ is ill-conditioned. Orthogonalizing through $\mathbf{S}^{-1/2}$ exposes small overlap eigenvalues and lets the code discard directions that would amplify roundoff or sampling noise.
3. **Variational subspace.** The orthonormalized vectors span the same retained subspace as the raw Chebyshev vectors. Therefore the Ritz eigenvalues are the same as those from the generalized problem on that retained subspace.
4. **Amplitude thresholding.** The implementation's `basis_threshold` acts on the raw generated vectors before orthonormalization. Linear recombination can reintroduce small amplitudes in the returned orthonormal vectors.
5. **Basis ordering.** The orthonormal basis is ordered according to the retained overlap eigenvectors, not by Chebyshev degree. Energies and subspace properties are invariant under unitary rotations of the orthonormal basis, but the rows of `basis_states` should not be read as time/order steps.

8 Code-to-theory correspondence

Theory object	Code expression	Meaning
B	<code>raw_basis_states</code>	raw Chebyshev state rows
S	<code>current_basis_states.conj() @ current_basis_states.T</code>	raw overlap
H_K	<code>current_basis_states.conj() @ (H.mat @ current_basis_states.T)</code>	raw projected Hamiltonian
U_r, Λ_r	<code>s_vecs[:, keep], s_vals[keep]</code>	retained overlap eigenspace
X	<code>s_vecs[:, keep] / sqrt(s_vals[keep])</code>	Lowdin transform
Q	<code>X.T @ current_basis_states</code>	returned orthonormal basis
H_{ortho}	<code>X.conj().T @ H_k @ X</code>	Hamiltonian in returned basis

9 References

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